Framework for Portfolio Allocation

Haibei Zhao

1 Overview

I propose a modular framework for adaptive portfolio allocation that separates prediction, risk modeling, structural break detection, and dynamic control. The key design principle is that reinforcement learning (RL) is used as an allocation controller on top of structured financial summaries, rather than as an end-to-end black box.

At each rebalancing time t, the system consists of four layers:

$$\begin{aligned}
\hat{\mu}_t &= \mathcal{M}_\mu(X_t), \\
\hat{\Sigma}_t &= \mathcal{M}_\Sigma(Y_t), \\
c_t &= \mathcal{M}_{\mathrm{CP}}(U_{1:t}), \\
a_t &= \pi_\theta(s_t),
\end{aligned}$$

where $\hat{\mu}_t$ is the forecasted cross-section of expected excess returns, $\hat{\Sigma}_t$ is the estimated conditional covariance matrix, c_t is a change-point/regime summary, and π_θ is the RL controller.

2 State, Action, and Portfolio Transition

Let N denote the number of assets, and let $w_t \in \mathbb{R}^N$ denote the portfolio weights at time t. The state s_t is constructed as a sufficient statistic of the financial environment and can be defined as

$$s_t = \big(\hat{\mu}_t, \hat{\Sigma}_t, c_t, w_{t-1}, \text{cost proxies}, \text{portfolio constraints}, \text{macro state}\big).$$

A direct action design would let the RL output target weights or weight changes. However, for interpretability and stability, our preferred design is an **optimizer-parameter controller**:

$$a_t = (\lambda_t, \tau_t, G_t, d_t),$$

where λ_t is a time-varying risk-aversion parameter, τ_t is a turnover penalty or cap, G_t is a gross leverage limit, and d_t is a defensive-mode switch.

Conditional on a_t and the regime signal c_t, the actual portfolio is obtained from a constrained optimizer:

$$w_t = \arg\max_{w \in \mathcal{W}_t(a_t, c_t)} \Big(\hat{\mu}_t^\top w - \lambda_t\, w^\top \hat{\Sigma}_t w - \tau_t \|w - w_{t-1}\|_1\Big).$$

The feasible set $\mathcal{W}_t(a_t, c_t)$ may include long-only, gross-exposure, sector, and turnover constraints.

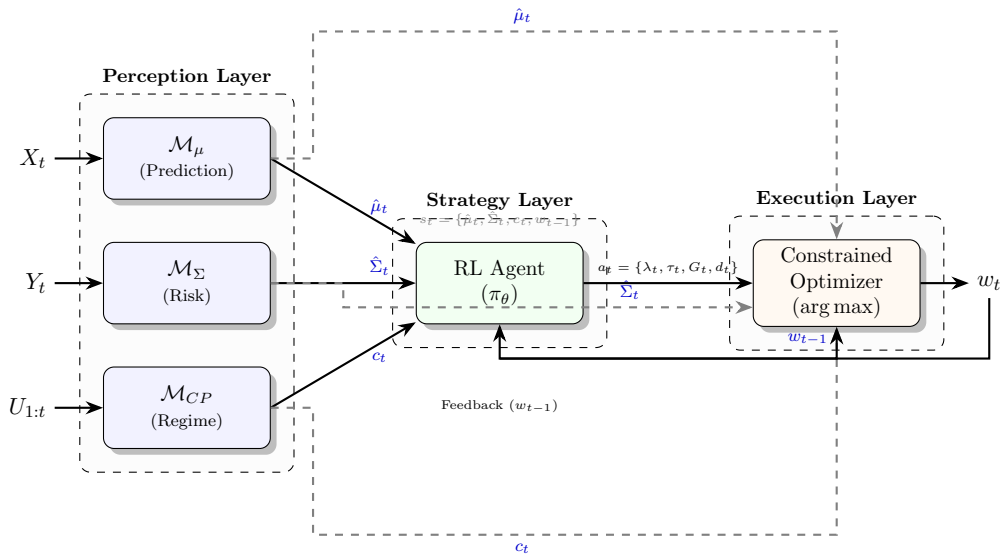

Figure 1: Decoupled Hierarchical Architecture

Table 1: Summary of RL Algorithm Suitability(Hyperparameter Controller)

Feature	**Best Algorithm**	**Reasoning for Portfolio Meta-Control**
Action Space	SAC / PPO	Naturally handles the continuous domain of (λ_t, τ_t) without the artifacts or resolution loss of discretization.
Stability	PPO	The trust-region/clipping mechanism prevents catastrophic shifts in risk aversion (λ_t), ensuring smoother transitions in portfolio behavior.
Robustness	SAC	Maximum entropy exploration prevents the policy from collapsing into a single deterministic (overfitted) value, favoring strategies that work across regimes.
Sample Efficiency	SAC	As an off-policy method, it enables intensive reuse of historical price trajectories within the Exo-MDP framework, speeding up convergence.

3 Prediction Layer for Expected Returns

The expected return layer produces

$$\hat{\mu}_{i,t} \approx \mathbb{E}_t[r_{i,t+1}], \qquad i = 1, \ldots, N.$$

Candidate models include:

(i) linear baselines ($\boxed{\text{Fama–MacBeth}}$, ridge, lasso, elastic net),

(ii) $\boxed{\text{tree-based methods (Random Forest, XGBoost, LightGBM)}}$,

(iii) neural networks (MLP),

(iv) structural asset-pricing decompositions such as IPCA-implied expected returns,

(v) ensembles of the above.

4 Risk Layer

The risk layer estimates

$$\hat{\Sigma}_t \approx \mathrm{Cov}_t(r_{t+1}).$$

Practical candidates include:

(i) $\boxed{\text{shrinkage covariance estimators}}$,

(ii) exponentially weighted moving average (EWMA) covariance,

(iii) factor covariance models

$$\hat{\Sigma}_t = B_t \hat{\Sigma}_{f,t} B_t^\top + D_t,$$

where B_t is a factor loading matrix, $\hat{\Sigma}_{f,t}$ is a factor covariance matrix, and D_t is idiosyncratic risk,

(iv) dynamic correlation models for smaller asset universes.

A factor-based covariance model is especially attractive when the prediction layer already includes factor or IPCA structure.

5 Change-Point / Regime Layer

The regime layer detects structural instability in market conditions and outputs a regime summary

$$c_t \in \mathcal{C}.$$

Possible outputs include a binary break indicator, a scalar alarm score, or a regime-posterior vector such as

$$c_t = \big(p_t^{\text{calm}}, p_t^{\text{volatile}}, p_t^{\text{crisis}}, \text{break score}_t, \text{time since break}\big).$$

Candidate methods include:

(i) online CUSUM / GLR / MOSUM procedures,

(ii) Bayesian online change-point detection,

(iii) hidden Markov or Markov-switching models as soft-regime alternatives.

For a first version, applying change-point detection to market- and factor-level indicators, rather than to each individual asset.

6 Reward Design

Let the realized net portfolio return be

$$R_{p,t+1}^{\text{net}} = w_t^\top r_{t+1} - \kappa_1 \|w_t - w_{t-1}\|_1 - \kappa_2 \|w_t - w_{t-1}\|_2^2.$$

A simple benchmark reward is

$$\boxed{r_t^{\text{RL}} = R_{p,t+1}^{\text{net}} - \lambda_r \, w_t^\top \hat{\Sigma}_t w_t.}$$

A more finance-oriented reward is based on utility, e.g.

$$r_t^{\text{RL}} = \log(1 + R_{p,t+1}^{\text{net}}) \quad \text{or} \quad r_t^{\text{RL}} = R_{p,t+1}^{\text{net}} - \frac{\gamma}{2} \operatorname{Var}_t(R_{p,t+1}).$$

A particularly appealing stepwise objective is the differential Sharpe ratio. Define exponentially weighted moments:

$$A_t = (1-\eta)A_{t-1} + \eta R_{p,t}, \qquad B_t = (1-\eta)B_{t-1} + \eta R_{p,t}^2,$$

and the associated Sharpe proxy

$$S_t = \frac{A_t}{\sqrt{B_t - A_t^2}}.$$

A common first-order incremental approximation yields the differential Sharpe reward

$$D_t \propto \frac{B_{t-1}(R_{p,t} - A_{t-1}) - \frac{1}{2}A_{t-1}(R_{p,t}^2 - B_{t-1})}{(B_{t-1} - A_{t-1}^2)^{3/2}}.$$

This reward encourages decisions that improve rolling risk-adjusted performance rather than merely increasing one-step return.

Table 2: Comparison of Stepwise Reward Metrics for RL Portfolio Control

Metric	**Focus**	**Ideal Use Case**
Diff. Sharpe Ratio (DSR)	Total Volatility	General alpha capture in stable markets
Diff. Sortino Ratio (DSortR)	Downside Deviation	Trend following with asymmetric risk
Diff. Information Ratio (DIR)	Tracking Error	Active management against a benchmark
Drawdown-Aversion	Path-dependent Risk	Tail risk control and capital preservation

Differential Sortino Ratio (DSortR) To focus on downside risk, we define the exponentially weighted downside second moment:

$$B_t^{down} = (1-\eta)B_{t-1}^{down} + \eta \min(0, R_{p,t})^2 \tag{1}$$

The DSortR reward at time t is then approximated as:

$$D_t^{Sortino} \approx \frac{B_{t-1}^{down}(R_{p,t} - A_{t-1}) - \frac{1}{2}A_{t-1}(\min(0, R_{p,t})^2 - B_{t-1}^{down})}{(B_{t-1}^{down})^{3/2}} \tag{2}$$

Differential Information Ratio (DIR) Let $R_t^{active} = w_t^\top r_t - R_{benchmark,t}$. Replacing $R_{p,t}$ with R_t^{active} in the DSR formulation yields:

$$D_t^{IR} \propto \frac{B_{t-1}^{active}(R_t^{active} - A_{t-1}^{active}) - \frac{1}{2}A_{t-1}^{active}((R_t^{active})^2 - B_{t-1}^{active})}{(B_{t-1}^{active} - (A_{t-1}^{active})^2)^{3/2}} \tag{3}$$

Drawdown-Aversion Reward We track the decaying peak of the portfolio value V_t:

$$M_t = \max(M_{t-1} \cdot e^{-\alpha}, V_t) \tag{4}$$

where α is a decay factor. The drawdown reward is defined as:

$$r_{DD,t} = R_{p,t}^{net} - \phi \cdot \max\left(0, \frac{M_t - V_t}{M_t} - \delta\right) \tag{5}$$

where δ is the tolerance threshold and ϕ is the penalty strength.

7 Initial Specification

Version 1:

(1) Expected return model: Fama-MacBeth, Tree-based methods(XGBoost/LightGBM/Random Forest) ,

(2) Risk model: shrinkage factor covariance estimators,

(3) Change-point model: online CUSUM on market/factor indicators,

(4) RL controller: SAC over low-dimensional optimizer parameters $a_t = (\lambda_t, \tau_t)$, where λ_t is a time-varying risk-aversion parameter, τ_t is a turnover penalty or cap, Conditional on a_t and the regime signal c_t, the actual portfolio is obtained from a constrained optimizer:

$$w_t = \arg\max_{w \in \mathcal{W}_t(a_t, c_t)} \left(\hat{\mu}_t^\top w - \lambda_t\, w^\top \hat{\Sigma}_t w - \tau_t \|w - w_{t-1}\|_1\right).$$

(5) Main reward: tbd

(6) Benchmark reward: net return minus risk minus turnover.

This design preserves interpretability, supports modular ablation studies, and keeps RL in a well-defined role as a sequential allocation controller.

8 Data and Sample Construction

To ensure the reproducibility of our RL-based allocation system, this section details the data universe, primary sources, and the synchronization logic used to construct the multi-layer input panel.

8.1 Investment Universe and Sample Period

The primary universe consists of historical constituents of the S&P 500 index. To mitigate survivorship bias, we utilize a dynamic membership list rather than the current index composition.

- **Sample Period:** January 2000 to December 2025.
- **Frequency:** Monthly rebalancing, aligned with the release of macroeconomic indicators and standard accounting reporting cycles.
- **Constituent Selection:** Stocks are included in the investable universe at month t only if they were members of the S&P 500 at the end of month $t-1$.

8.2 Data Sources and Integration

Data is retrieved from the Wharton Research Data Services (WRDS) platform, integrating three distinct modalities:

1. **Market Data (CRSP):** Monthly returns (ret), prices (prc), and shares outstanding ($shrout$) are sourced from the `crsp.msf` table. We adjust for delisting returns to prevent upward bias in performance metrics.
2. **Fundamental Data (Compustat):** Annual accounting items (e.g., Book Value, Operating Income) are retrieved from `comp.funda`. To avoid look-ahead bias, we apply a conservative six-month minimum reporting lag.
3. **Factor Data (Fama-French):** The Fama-French 3-factor and 5-factor series, along with the risk-free rate, are used to anchor the *Prediction Layer* and calculate excess returns.

8.3 Feature Engineering and Pre-processing

The "Perception Layer" (State s_t) requires a synchronized panel. We implement the following pre-processing steps:

- **CCM Linking:** Compustat *gvkey* identifiers are mapped to CRSP *permno* using the `crsp.ccmxpf_lnkhist` table with valid link filters (`LU`, `LC`).
- **Normalization:** Cross-sectional features are winsorized at the 1% and 99% levels to manage outliers and then Z-scored to ensure stability for the RL controller.
- **Risk Layer Alignment:** Realized return windows for the *Shrinkage Covariance* estimator are strictly partitioned from the target prediction window to ensure a valid walk-forward test.

9 Evaluation

The system should be evaluated in a fully walk-forward setting with nested training/validation/testing splits to avoid look-ahead bias. Performance should be compared against:

(i) static mean-variance allocation,

(ii) robust allocation without RL,

(iii) RL without change-point inputs,

(iv) RL with direct weights versus RL with optimizer-parameter control.